Midterm Project Report: Vitamin Deficiency Disease Analysis

1. Introduction & Literature Review

Vitamin deficiencies are a significant global health concern, contributing to a wide range of diseases and adverse health outcomes. Understanding the relationship between nutritional biomarkers and disease diagnosis is essential for early detection and prevention. This project explores the use of machine learning techniques to classify vitamin deficiency-related diseases based on patient data.

Current research in healthcare analytics increasingly leverages machine learning models such as Random Forests and gradient boosting methods to improve diagnostic accuracy. Studies have shown that ensemble learning methods can effectively capture complex, non-linear relationships in medical data. Additionally, organizations in healthcare and nutrition analytics utilize predictive models to assist clinicians in identifying deficiencies earlier.

However, challenges remain, particularly regarding imbalanced datasets and limited feature representation. Many models struggle with minority class prediction, which can reduce reliability in real-world applications. This project addresses these challenges by evaluating multiple models and analyzing their performance in a multi-class classification setting.

2. Problem Definition

The objective of this project is to develop a machine learning model that can accurately classify vitamin deficiency diseases based on patient features.

Inputs (Features): Numeric patient attributes including demographic data and nutritional/laboratory measurements such as age, BMI, and vitamin intake percentages.
Output (Target): Disease category representing a specific vitamin deficiency condition.

This is a multi-class classification problem where the goal is to assign each patient record to one of several disease categories.

3. Approach, Materials, and Methods

Dataset: The dataset used is 'vitamin_deficiency_disease_dataset_20260123.csv' sourced from Kaggle. It contains patient-level data with both demographic and nutritional variables.

Data Preprocessing: The data underwent cleaning, including handling missing values and removing problematic columns. Feature engineering and transformation were applied to prepare the dataset for modeling.

Exploratory Data Analysis (EDA): Visualizations included disease distribution plots, histograms of features, correlation heatmaps, and comparisons between disease and dietary patterns.

Models Used:
- Random Forest Classifier
- Linear Regression (baseline)
- XGBoost Classifier

Model Selection Rationale: Random Forest was chosen for its robustness and ability to capture non-linear relationships. Linear Regression was used as a baseline to demonstrate performance differences, while XGBoost was included for its strong predictive capability.

Tools and Libraries: Python, pandas, numpy, matplotlib, seaborn, scikit-learn, and xgboost.

4. Results

The models were evaluated using a stratified 20% test split. Performance metrics included accuracy, classification reports, and confusion matrices.

Random Forest Classifier: Demonstrated solid overall performance with higher precision and recall for majority classes, while minority classes showed lower recall.

Linear Regression: Performed poorly compared to classification models due to its inability to properly model discrete outputs.

XGBoost Classifier: Achieved competitive performance and often outperformed the Random Forest model due to its boosting approach.

Overall, the Random Forest model provided a strong balance between interpretability and performance.

5. Discussion

The results indicate that ensemble models are effective for this classification task, though performance is impacted by class imbalance. Minority classes consistently showed lower recall, suggesting that the model struggles to identify less frequent conditions.

Parameter tuning and class weighting improved results, but limitations remain due to dataset imbalance and limited feature diversity. Misclassifications were most common between similar deficiency types, indicating overlapping feature patterns.

Compared to typical state-of-the-art approaches, XGBoost performed competitively, though additional tuning could further improve results. The Random Forest model was selected due to its balance of performance and interpretability.

6. Conclusion

This project successfully applied machine learning techniques to classify vitamin deficiency diseases using patient data. The Random Forest and XGBoost models demonstrated strong performance, with ensemble methods proving effective for this task.

Key learnings include the importance of handling imbalanced datasets, the value of feature engineering, and the performance advantages of classification-specific models.

Future improvements include enhancing class balance, incorporating categorical features, optimizing hyperparameters, and exploring advanced ensemble techniques.